Dun Yuan, Mr.

dun.yuan@mail.mcgill.ca   Dun Yuan   +1(438)493-2343

Education

2021 – Present **Ph.D. Computer Science,** McGill University, Montreal, Canada
Supervisor: Dr. Xue Liu

2020 – 2021 **M.Sc. Computer Science,** Imperial College London, London, United Kingdom

2017 – 2017 **Fall Semester Exchange Program,** Columbia University, New York, United States

2015 – 2019 **Bachelor of Engineering in Automation,** Xi'an Jiaotong University, Xi'an, China

Employment History

2021 – 2023 **Research Intern,** Samsung AI Center - Montreal, Montreal, Canada

Research Publications

1. H. Zhou, C. Hu, D. Yuan, *et al.*, "Generative ai as a service in 6g edge-cloud: Generation task offloading by in-context learning," *arXiv preprint arXiv:2408.02549*, 2024.
2. H. Zhou, C. Hu, D. Yuan, *et al.*, "Large language model (llm)-enabled in-context learning for wireless network optimization: A case study of power control," *arXiv preprint arXiv:2408.00214*, 2024.
3. H. Zhou, C. Hu, Y. Yuan, *et al.*, "Large language model (llm) for telecommunications: A comprehensive survey on principles, key techniques, and opportunities," *arXiv preprint arXiv:2405.10825*, 2024.
4. D. Yuan, E. Hossain, D. Wu, X. Liu, and G. Dudek, "Realizing xr applications using 5g-based 3d holographic communication and mobile edge computing," *arXiv preprint arXiv:2310.03908*, 2023.
5. D. Yuan, Y. Nam, A. Feriani, *et al.*, "Mixed-variable pso with fairness on multi-objective field data replication in wireless networks," in *ICC 2023-IEEE International Conference on Communications*, IEEE, 2023, pp. 3720–3725.
6. D. Yuan and L. Liu, "Design and implementation of vehicle chassis detection system based on multi-sensor fusion technology," in *2019 4th International Conference on Electromechanical Control Technology and Transportation (ICECTT)*, IEEE, 2019, pp. 113–116.

Research interests

Artificial Intelligence (AI)
• Machine learning methods and techniques, including supervised learning, unsupervised learning, and reinforcement learning (RL)
• Neural network architectures: convolutional neural networks (CNNs), recurrent neural networks (RNNs), and graph neural networks (GNNs)
• Natural language processing (NLP) models and applications

Large Language Models (LLMs)
• Domain-specific LLMs training and fine-tuning
• Optimizing LLMs using reinforcement learning from human feedback (RLHF)
• Detecting and mitigating hallucinations in LLMs
• Retrieval-augmented generation (RAG) frameworks with diverse knowledge databases
• Document-based text summarization and question answering with LLMs

Research interests (continued)

- **Web3.0**
  - Cryptography in distributed systems and decentralized networks
  - Incorporating distributed systems with AI techniques
  - Design of decentralized storage and computing systems

- **Computer Networks**
  - Telecommunication structures of 5G and beyond
  - Resource allocation in computer networks with mathematical optimization methods
  - Enhancing wireless network performance with RL techniques

Skills

Languages	English and Mandarin Chinese
Coding	Java, C, C++, C# Python, LaTeX, …

Awards

2022 – 2024	**Graduate Excellence Awards**, McGill University
2016 – 2017	**Second Prize**, Robocon Contest
2016 – 2016	**First Prize**, China Undergraduate Mathematical Contest in Modeling